

$$\mathbf{e}_{t,k}^X = \mathbf{P}^X \mathbf{W}_{t,k}^X \left( \mathbf{Y}_{t,k}^X - \hat{\mathbf{X}}_{t,k}^1 \right),$$

$$v_{\text{FBFM}}^X = v_{\theta}^X + \lambda_{\tau_k^X}^X \left( \mathbf{J}_{t,k}^X \right)^{\top} \mathbf{e}_{t,k}^X$$